

AUTOCAD CORE - RAG STRUCTURED DOCUMENT

Concept: Database

Definition: Central container representing a DWG file. Owns all DBObjects.

Key Properties: BlockTableId, LayerTableId, TransactionManager

Access Pattern: ObjectId -> Transaction -> DBObject

Invariant: All DBObjects belong to exactly one Database

```
Access ModelSpace:
var bt = (BlockTable)tr.GetObject(db.BlockTableId, OpenMode.ForRead);
var ms = (BlockTableRecord)tr.GetObject(bt[BlockTableRecord.ModelSpace], OpenMode.ForWrite);
```

Concept: Transaction

Definition: Unit of work ensuring atomic read/write.

Rules:

- Must wrap all DB access
- Commit persists changes
- Dispose without commit = rollback

```
using (var tr = db.TransactionManager.StartTransaction())
{
    // read/write
    tr.Commit();
}
```

Concept: Block Definition

Definition: BlockTableRecord storing reusable geometry.

Properties: Name, Origin, Entities

Invariant: No transform applied here

Concept: Block Reference

Definition: Instance of BlockDefinition.

Properties: Position, Scale, Rotation

Invariant: Transform applied at reference level

```
var br = new BlockReference(pt, blockDefId);
```

Concept: Attribute System

Definition: Key-value metadata attached to blocks

Flow:

1. AttributeDefinition in Block
2. AttributeReference in BlockReference
3. Sync required if definition changes

Concept: Document

Definition: Runtime wrapper of Database + Editor

Manager: Application.DocumentManager

Constraint: Must lock before write

```
using (doc.LockDocument()) { }
```

Concept: Editor

Definition: UI interaction layer

Capabilities: Selection, Prompt, Input

Operation: Insert Block

Steps:

1. Open BlockTable
2. Get definition
3. Create BlockReference
4. Append to ModelSpace

Operation: Deep Clone

Definition: Copy objects across DBs

Tool: IdMapping

```
db.WblockCloneObjects(ids, targetId, mapping, DuplicateRecordCloning.Ignore, false);
```

Performance Rules

- Batch operations in single transaction
- Prefer ForRead over ForWrite
- Cache ObjectIds
- Avoid Editor in loops

Anti-patterns

- Transaction per entity
- Global ForWrite access
- Missing DocumentLock
- Repeated symbol table lookup by name

Threading

[Unverified] AutoCAD DB is not thread-safe. Use main thread for DB ops.

ODA Mapping

AutoCAD: AcDbDatabase | ODA: OdDbDatabase

AutoCAD: ObjectId | ODA: OdDbObjectId

Concept parity: ~95%